

# Static Gesture Classification — Run Report

Generated 2026-05-07 22:24:08

## Data paths

|                      |                                                                   |
|----------------------|-------------------------------------------------------------------|
| Training data path 1 | /home/jestin/ThesisRepo/ML/NewTestData/Alan/Static · 35 trials    |
| Training data path 2 | /home/jestin/ThesisRepo/ML/NewTestData/Alex/Static · 35 trials    |
| Training data path 3 | /home/jestin/ThesisRepo/ML/NewTestData/Georgia/Static · 35 trials |
| Training data path 4 | /home/jestin/ThesisRepo/ML/NewTestData/Henry/Static · 35 trials   |
| Training data path 5 | /home/jestin/ThesisRepo/ML/NewTestData/Jestin/Static · 36 trials  |
| Training data path 6 | /home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Static · 30 trials |
| Training data path 7 | /home/jestin/ThesisRepo/ML/NewTestData/Josh/Static · 35 trials    |
| Training data path 8 | /home/jestin/ThesisRepo/ML/NewTestData/Nat/Static · 37 trials     |
| Training data path 9 | /home/jestin/ThesisRepo/ML/NewTestData/Mansh/Static · 36 trials   |
| Total training paths | 9 · 314 trials total                                              |
| External test root   | /home/jestin/ThesisRepo/ML/NewTestData/Marcus/Static              |
| Report output dir    | /home/jestin/ThesisRepo/ML/saved_reports                          |

## Sensor selection

|                |                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| Hands          | left, right                                                                                                         |
| Segments       | palm_mid, thumb, index, middle, ring, pinky, wrist                                                                  |
| Modalities     | YPR, Accel, Flex                                                                                                    |
| Sensor columns | 176 channels                                                                                                        |
| Classes        | 7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman |

## Preprocessing, features & split

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Resample steps              | 90                                         |
| Butterworth                 | on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz  |
| Normalisation               | minmax                                     |
| Feature mode                | stats → feature dim 1408                   |
| Test size · seed · CV folds | 0.2 · random_state=42 · 10-fold stratified |
| Sample counts               | train (post-aug)=502 · test=63             |

## Augmentation (training only)

|                        |                        |
|------------------------|------------------------|
| Enabled                | yes                    |
| Copies per train trial | 1                      |
| Jitter                 | off · sigma=0.01       |
| Amplitude scale        | on · range=(0.8, 1.2)  |
| Time warp              | off · range=(0.9, 1.1) |

## Classifiers — cross-validation & hold-out test

| Algorithm                       | CV mean (10-fold) | CV std | Hold-out test acc |
|---------------------------------|-------------------|--------|-------------------|
| Logistic Regression             | 1.0000            | 0.0000 | 0.9524            |
| SVM (RBF)                       | 1.0000            | 0.0000 | 0.9365            |
| SVM (Linear)                    | 1.0000            | 0.0000 | 0.9365            |
| Random Forest                   | 1.0000            | 0.0000 | 0.9365            |
| Voting Soft (SVM+RF+KNN+LogReg) | 1.0000            | 0.0000 | 0.9365            |
| Stacking (SVM+RF+KNN+LogReg)    | 1.0000            | 0.0000 | 0.9365            |
| KNN                             | 0.9562            | 0.0250 | 0.8889            |
| LDA                             | 0.9542            | 0.0347 | 0.2540            |

## Per-class hold-out F1

| Class              | Logistic Regression | SVM (RBF) | SVM (Linear) | Random Forest | Voting (soft) | Stacking | KNN   | LDA   | Support |
|--------------------|---------------------|-----------|--------------|---------------|---------------|----------|-------|-------|---------|
| Double_Closed_Fist | 1.000               | 1.000     | 1.000        | 1.000         | 1.000         | 1.000    | 1.000 | 0.308 | 10      |
| Double_Nothing     | 0.933               | 0.857     | 0.857        | 0.857         | 0.857         | 0.857    | 0.857 | 0.133 | 8       |
| Double_Okay        | 0.947               | 0.900     | 0.900        | 1.000         | 0.900         | 0.900    | 0.737 | 0.118 | 9       |
| Double_Open_Palm   | 0.941               | 0.941     | 0.941        | 1.000         | 0.941         | 0.941    | 0.778 | 0.167 | 9       |
| Double_Phone       | 0.941               | 0.941     | 0.941        | 0.875         | 0.941         | 0.941    | 0.941 | 0.125 | 9       |
| Double_Pistol      | 1.000               | 1.000     | 1.000        | 0.900         | 1.000         | 1.000    | 0.947 | 0.476 | 9       |
| Double_Spiderman   | 0.900               | 0.900     | 0.900        | 0.900         | 0.900         | 0.900    | 0.947 | 0.316 | 9       |
| macro avg          | 0.952               | 0.934     | 0.934        | 0.933         | 0.934         | 0.934    | 0.887 | 0.235 | 63      |

## External test (NewTestData)

| Algorithm     | External test acc |
|---------------|-------------------|
| Random Forest | 1.0000            |

